MIL-D-2352C

11 April 1968
SUPERSEDING
MIL-D-2352B
29 September 1964

MILITARY SPECIFICATION

DOUGH ROUNDING MACHINE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an electrically operated dough rounding machine.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-641 - Motor, Alternating Current (Integral Horse Power).
QQ-I-652 - Iron Casting, Gray.

MILITARY

MIL-B-2316 - Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Interference Characteristics - Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

FSC 7320

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Electrical Construction Materials List

Standard UL 73-1967: Motor Operated Appliances

(Requests for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L. I., New York 11746, or 1655 Scott Boulevard, Santa Clara, California 95050.)

BAKING INDUSTRY SANITATION STANDARDS COMMITTEE (BISSC)

Sanitation Standard No. 8: Dividers, Rounders and Bun Machines

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N. Y. 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standard A 167-63 - Corrosion-Resisting Chromium-Nickel Steel Plate, and Strip.

ASTM Standard A 176-63 - Corrosion-Resisting Chromium Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19106.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

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3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample dough rounding machine shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standard product.- Dough rounding machines delivered under this specification shall be the manufacturers' current product, except for any changes necessary to comply with specification requirements.

3.3 Standards.- Dough rounding machines shall conform to the requirements of BISSC Standard No. 8. Electrical components shall conform to the applicable requirements of Standard UL 73-1967.

3.3.1 Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dough rounding machine he proposes to furnish under this specification meets the applicable requirements of BISSC Standard No. 8 and all electrical components meet the applicable requirements of UL 73-1967.

3.3.1.1 UL.- Acceptable evidence of meeting the requirements of the UL shall be the UL label, listing in the UL "Electrical Construction Materials List" or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electrical components offered have been tested and conform to the applicable requirements of Standard UL 73-1967.

3.3.1.2 BISSC.- Acceptable evidence of meeting the requirements of BISSC Seal of Approval, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the dough rounding machine offered has been tested and conforms to the applicable requirements of BISSC Standard No. 8.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dough rounding machine.

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* 3.4.1 Corrosion-resisting steel.- Corrosion-resisting steel used for food-contact surfaces shall conform to any of the type 300 series specified in ASTM A 167-63. When corrosion-resisting steel is used for other surfaces, the chemical compositions of any of the type 300 series of ASTM A 167-63, or any of the type 400 series of ASTM A 176-63 shall be used.

3.4.2 Cast iron.- Cast iron shall conform to class 30, or better, of QQ-I-652.

3.4.3 Motor.- The motor shall conform to type II, totally inclosed of CC-M-641. The motor shall have ample power to operate the rounder continuously at the rate specified in 3.5 without exceeding the allowable temperature rise. Unless otherwise specified (see 6.2), the motor shall be designed to operate on a nominal 208-volt, 60 hertz, 3-phase alternating current (a.c.) system.

* 3.5 Design.- The dough rounding machine shall be either the rotating cone or umbrella type, with an adjustable dough race designed to provide the desired rounding effect, without the need for special coatings on the spiral race. The machine shall be capable of shaping the dough, without damage, into ball-like forms, without pilling. The machine shall be designed to round pieces of straight or sponge dough weighing from 8 to 36 ounces. When tested as specified in 4.4.2, the machine shall be capable of handling 24 ounce dough pieces at a minimum rate of 60 pieces per minute. When tested as specified in 4.4.1, all components of the machine shall operate smoothly without excessive vibration or noise.

3.6 Construction.-

* 3.6.1 Base.- The base of each machine shall be constructed of cast iron specified in 3.4.2 or fabricated of steel. The base may be a fully inclosed type, housing the motor and driving mechanism, or an open type with integral legs. Provisions shall be made for anchoring the machine to the floor.

* 3.6.2 Bowl or drum.- The machine shall have an inverted cone bowl, with an inside climbing spiral dough race, or an umbrella type drum with outside climbing spiral dough race. Both types of machines shall have the rotating section mounted on a rigid central shaft or spindle to insure accurate alignment and concentricity of the bowl or drum, and dough race. The shaft of spindle shall be provided with ball or roller bearings.

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* 3.6.3 Dough races and chutes.- The dough race shall be mounted on adjustable supports designed to permit adjustment for the required clearance between the race and rotating bowl or drum to avoid scrapping or pilling of dough. Each race shall terminate with a discharge chute arranged to deliver the rounded dough pieces to the proofer. An adjustable duster shall be provided over the discharge chute with a quick-acting clutch or shutoff to terminate flour delivery when not required. The intake and discharge chutes shall be located to facilitate delivery from the divider to the proofer and shall be located within easy reach of the operator from one position. When a special chute height is required, the height shall be specified (see 6.2). A sub-base may be furnished, where necessary, to meet the above conditions.

3.6.4 Switches.- The switch shall be of the ON-OFF pushbutton type, with thermal overload and low-voltage protection. The switch shall be designed to permit the use of an auxiliary push button on the panel at the scale stand. The auxiliary push-button switch shall be supplied with each machine.

3.6.5 Lubrication.- Provision shall be made for lubrication at all points requiring it. Pressure gun lubrication fittings shall be provided at readily accessible points for all bearings requiring grease lubrication. At time of delivery, the machine shall be lubricated with the manufacturer's standard lubricant, except permanently lubricated ball or roller bearings.

3.6.5.1 Lubricating chart.- The supplier shall affix to each machine a corrosion-resistant metal lubricating chart indicating:

(a) All points of the equipment requiring lubrication.
(b) Grade of lubricant required.
(c) Time interval when lubrication should be accomplished.

* 3.6.6 Electromagnetic compatibility.- Equipment procured under this specification for Army use, and, when specified (see 6.2) for Navy and Air Force use, shall conform to class IIB of MIL-STD-461.

3.7 Finish.- All exterior surfaces of the machine not coming in contact with the dough, except corrosion-resisting steel, shall be given a white enamel finish.

3.8 Marking for identification.- Each machine shall be marked for identification on a corrosion-resisting steel plate with the manufacturer's name or trade name and model number.

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* 3.9 Workmanship.- The finished machine shall be free from defects including marred surfaces, dents, punctures, fractures, rough edges, burrs, slivers. All welds shall be smooth and uniform, completely fused, without undercuts, cracks, burnholes or fractures. All fasteners shall be tight. No component shall be missing, or malformed. All components shall be carefully assembled, properly adjusted, and aligned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required it shall be examined for the defects in table I and dimensions specified, and tested in accordance with 4.4.1, 4.4.2 unless waived (see 6.3) and 4.4.3, if required (see 6.2).

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The inspection lot shall be the number of dough rounding machines offered for acceptance at one time. The sample unit for this inspection shall be one completely fabricated dough rounding machine.

* 4.3.2.1 Visual examination.- Examination of the end item shall be made for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished as specified		X
			X
Construction and workmanship (general applicable to all components and assemblies)	Any component loose		
	Bowl drum or dough race not concentric about drive shaft	X	
	Sharp burr, sliver, or splinter	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed or otherwise impaired.	X	
Base	No provision for anchoring machine to floor	X	
Dough races and chutes	Adjustable duster not provided	X	
Switches	Auxiliary push button not provided	X	
Lubrication	Not specified type lubrication fittings	X	
	Lubrication fittings not readily accessible for servicing		X
	Machine not lubricated at time of delivery		X
Welding and brazing where required	Missing, incomplete, burn holes, cracked fractured	X	
	Undercut, not uniform, scale, or flux deposits not removed		X
Screws, nuts and bolts	Any threaded fasteners missing, broken, stripped, or fractured	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Electrical assembly	Wiring cut, not properly joined, loose at terminal, or not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain, or required - insulation stripped from wiring		X
Identification marking	Missing, incomplete, not legible, not specified type		X

4.3.2.2 Dimensional examination.- Dimensional examination shall be made of the end item to determine compliance with this specification. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- Each production item shall be tested in accordance with 4.4.1. When no preproduction sample is required, the initial item from each lot shall also be tested as specified in 4.4.2, unless waived (see 6.3), and 4.4.3, if required (see 6.2).

4.3.3 Examination of preparation for delivery.- Examination of preservation, packaging, packing and marking shall be in accordance with MIL-B-2316.

4.3.4 Standards compliance.- Proof of compliance with UL and BISSC requirements of 3.3 shall be available to the Government representative.

4.4 Tests.-

4.4.1 Operation test.- The dough rounding machine shall be connected to an appropriate power source and operated without load for a minimum of 30 minutes to determine if all components operate efficiently, and to determine compliance with the lack of excessive noise and excessive vibration requirements of 3.5.

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4.4.2 Functional test.- Bread dough shall be processed in the dough rounding machine and the time rate of production, without pilling or damage to the rounds, checked to determine compliance with 3.5 (see 6.3).

* 4.4.3 Electromagnetic compatibility test.- When applicable, the preproduction sample, or initial unit of production, equipped for the reduction of radio interference in accordance with 3.6.6 shall be tested by the supplier in accordance with test methods CE03, CE05, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The supplier shall furnish the contracting officer, for approval, the Interference Control Plan, the EMI/EMC Test Plan and the electromagnetic compatibility test report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production, as applicable, shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use.- The dough rounding machines covered by this specification are intended for use in garrison bakeries for loaf rounding.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) First article.- When preproduction sample is not required (see 3.1).

(c) Electrical requirements if other than specified (see 3.4.3 and 3.6.4).

(d) Chute height (see 3.6.3).

(e) When radio interference suppression is required for Navy and Air Force use (see 3.6.6).

(f) Selection of the applicable levels of preservation and packaging and packing (see 5.1).

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6.3 Waiver of functional test.- The functional test specified in 4.4.2 may be waived provided that the supplier furnishes copies of certified test data covering previous tests of an identical machine, to the contracting officer. The test data must indicate compliance with the specified test conditions and must have been obtained as a result of tests conducted within the past 4 years on an identical dough rounding machine furnished on a Government contract. In addition to test data, a certified statement will be required.

* 6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0242

Review activities:

Army - EL, MD
Navy - MC, SA, YD
Air Force - 84, 85

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-D-2352C - DOUGH ROUNDING MACHINE

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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